Corrected smartphone thermal model

The corrected Icepak mesh and steady-state solve completed. The saved battery maximum is **54.3 C** and processor package maximum is **74.5 C** at 5.5 W total prescribed heat and 25 C ambient.

Run evidence	Recorded result
Mesh	671,368 cells; 697,776 nodes
Energy residual	6.8952×10^{-9}; target 1×10^{-8}
Solver completion	Normal Completion; 18 Oct 2026, 20:20:29 PDT
Temperature limiting	No temperature-limiter messages in the corrected transcript

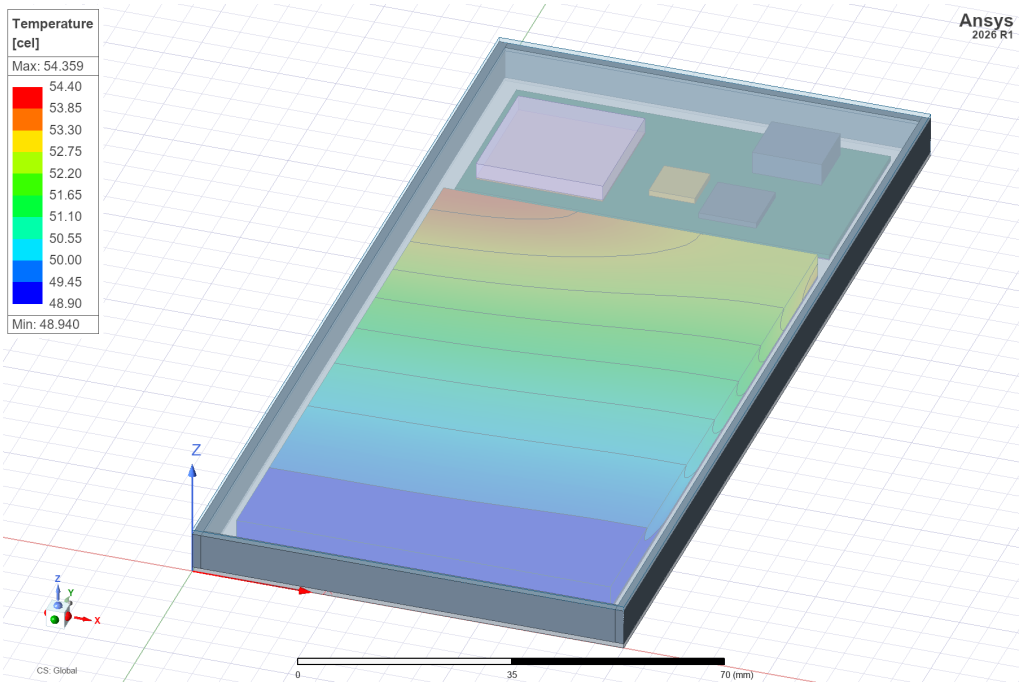


Figure 1. Fresh native Icepak battery surface temperature display. Legend extrema are surface-display values; the reported component maximum comes from the saved volume summary.

Status: completed demonstration with provisional results. The detailed transcript contains an initialization error. Global heat balance, mesh independence and physical validation remain open. This report documents those limits alongside the results.

Model scope and physical assembly

A simplified 75 x 160 x 8 mm assembly represents the enclosure, display, battery, PCB, packages and thermal interfaces. It is an illustrative model and is not a model of a named commercial phone.

Illustrative 75 × 160 × 8 mm assembly

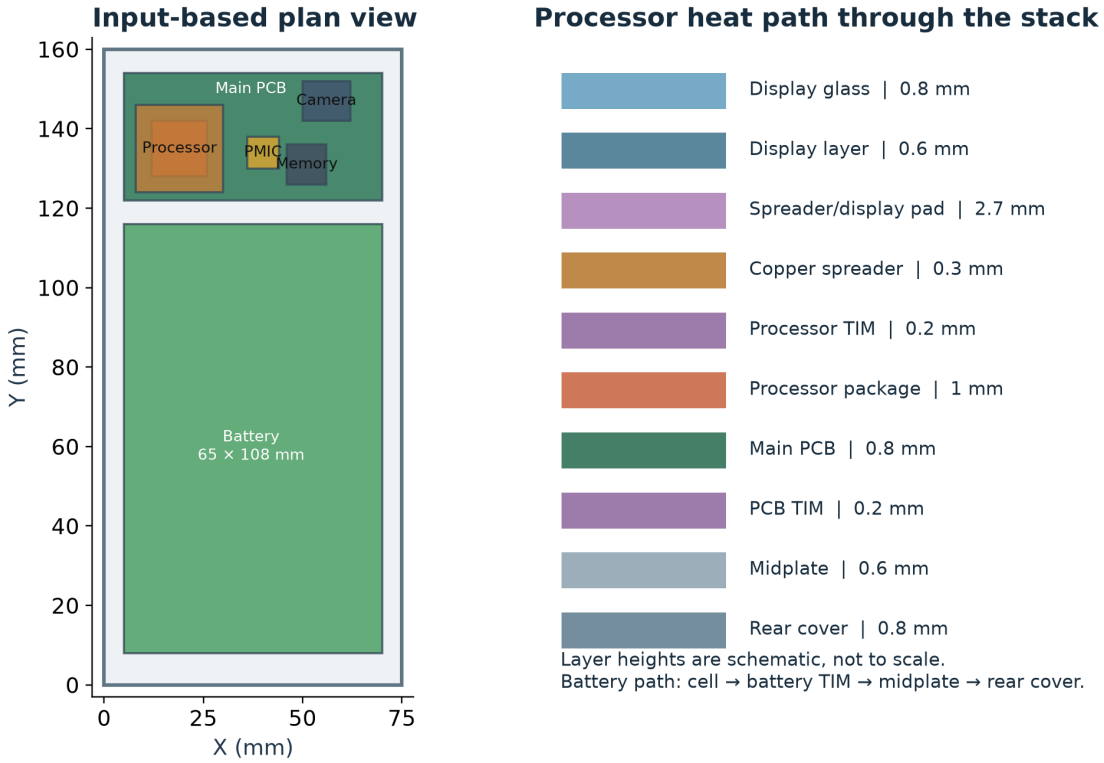


Figure 2. Schematic derived from the input dimensions. It explains the component layout and processor-to-display stack; this figure is not a solver contour.

The battery is an effective thermal solid with a fixed 0.8 W source. The PCB uses effective orthotropic conductivity. Individual traces, vias, detailed semiconductor junctions and battery electrochemistry are not resolved.

Touching solids use ideal bonded contact. Explicit TIM solids represent selected interfaces. A 0.2 mm air gap remains above the camera, while its base conducts into the PCB.

Complete component inventory

Geometry coordinates and dimensions are in mm. The table includes all 19 physical solids. The enclosing air Region is separate. Inventory source: original_inputs_unchanged.json.

Part	Origin X/Y/Z	Size X x Y x Z	Material	W
Rear Cover	0 / 0 / 0	75 x 160 x 0.8	Aluminium	0
Midplate	1.5 / 1.5 / 0.8	72 x 157 x 0.6	Aluminium	0
Frame Left	0 / 0 / 0.8	1.5 x 160 x 6.4	Aluminium	0
Frame Right	73.5 / 0 / 0.8	1.5 x 160 x 6.4	Aluminium	0
Frame Bottom	1.5 / 0 / 0.8	72 x 1.5 x 6.4	Aluminium	0
Frame Top	1.5 / 158.5 / 0.8	72 x 1.5 x 6.4	Aluminium	0
Display Glass	0 / 0 / 7.2	75 x 160 x 0.8	Glass	0
Display Layer	1.5 / 1.5 / 6.6	72 x 157 x 0.6	Display	1.2
Battery TIM	5 / 8 / 1.4	65 x 108 x 0.5	TIM	0
Battery Cell	5 / 8 / 1.9	65 x 108 x 4.2	Battery effective	0.8
PCB TIM	5 / 122 / 1.4	65 x 32 x 0.2	TIM	0
Main PCB	5 / 122 / 1.6	65 x 32 x 0.8	PCB effective	0
Processor Package	12 / 128 / 2.4	14 x 14 x 1	Package	2.5
Processor TIM	12 / 128 / 3.4	14 x 14 x 0.2	TIM	0
Copper Spreader	8 / 124 / 3.6	22 x 22 x 0.3	Copper	0
Spreader Display Pad	8 / 124 / 3.9	22 x 22 x 2.7	Processor pad	0
PMIC Package	36 / 130 / 2.4	8 x 8 x 1	Package	0.5
Memory Package	46 / 126 / 2.4	10 x 10 x 1	Package	0.3
Camera Module	50 / 142 / 2.4	12 x 10 x 4	Camera	0.2

The generated grid_mapping lists these 19 names, including Battery TIM, PCB TIM, Processor TIM and Copper Spreader. That confirms their inclusion in the mesh mapping; individual layer quality still requires local inspection.

Materials and prescribed heat

All properties and power losses are illustrative inputs. Conductivity uses W/(m K), density kg/m3, and specific heat J/(kg K). Conductivity triples are global X/Y/Z components.

Material	Conductivity	Density	Specific heat
Aluminium	160	2700	900
Battery effective	15 / 15 / 0.8	2300	1000
Camera	0.5	1800	900
Copper	390	8960	385
Display	0.5	1800	1000
Glass	1	2500	800
PCB effective	15 / 15 / 0.35	1900	1000
Package	2	1800	900
Processor pad	3	2000	1000
TIM	1.2	1800	1000

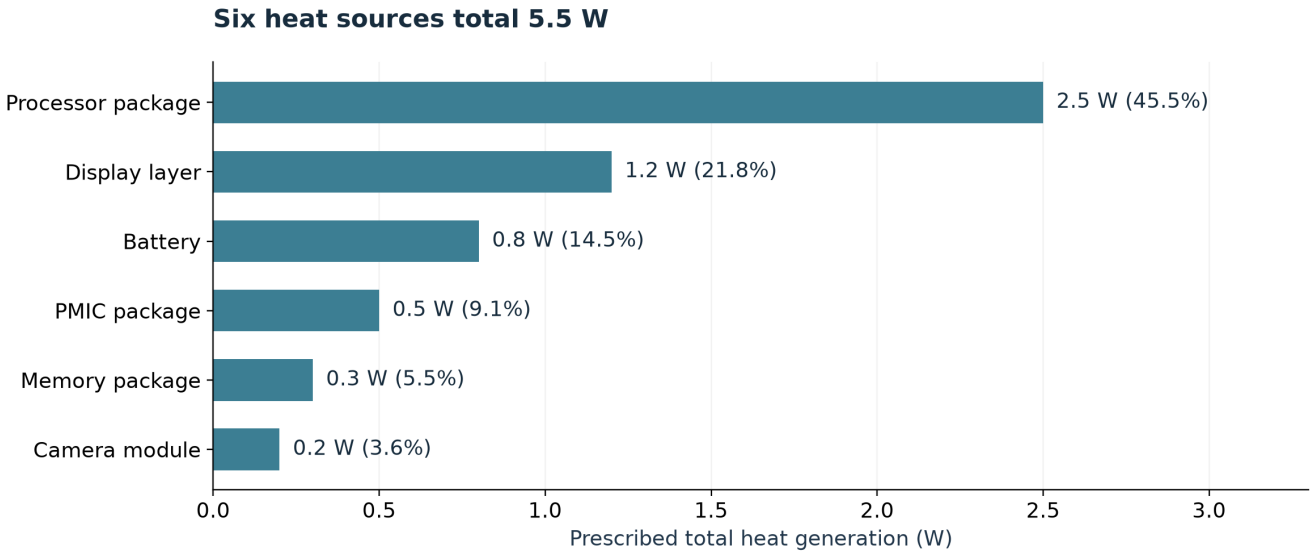


Figure 3. Six fixed heat inputs sum to 5.5 W. Processor dissipation is the largest individual source.

Density and heat capacity are recorded for future transient work. This completed run is steady state and does not predict heating duration, battery discharge or a mission profile.

Thermal physics and cooling boundary

Item	Applied model
Heat transfer	Steady conduction in solids and internal air
External cooling	Uniform prescribed h = 7 W/(m2 K) at 25 C
Cooling assignment	18 actual exterior solid faces; area = 0.02776 m2
Airflow / gravity / radiation	Disabled; h is prescribed, not calculated from airflow
Contacts	Ideal bonded contact where solids touch; selected explicit TIM layers
Unmodeled effects	Hand contact, protective case, throttling, charging dynamics and battery chemistry

The enclosure removes heat through the rear cover, frame and display glass. The processor also has an upward path through TIM, copper spreader and pad to the display, plus a downward path through the PCB and midplate.

Repair to the previous setup: External_Convection was reassigned from the nonmeshed Region boundary to the actual exposed physical faces. The live assignment and generated mesh mapping contain the same 18 face IDs.

```
Exterior face IDs:
36, 37, 38, 39, 40, 93, 94, 95, 121, 123, 124,
149, 179, 203, 205, 206, 207, 208
```

The uniform h assumption is an input requiring confirmation. It is not a prediction of natural convection. No new cooling scenarios were solved while creating this report.

Generated mesh and thickness resolution

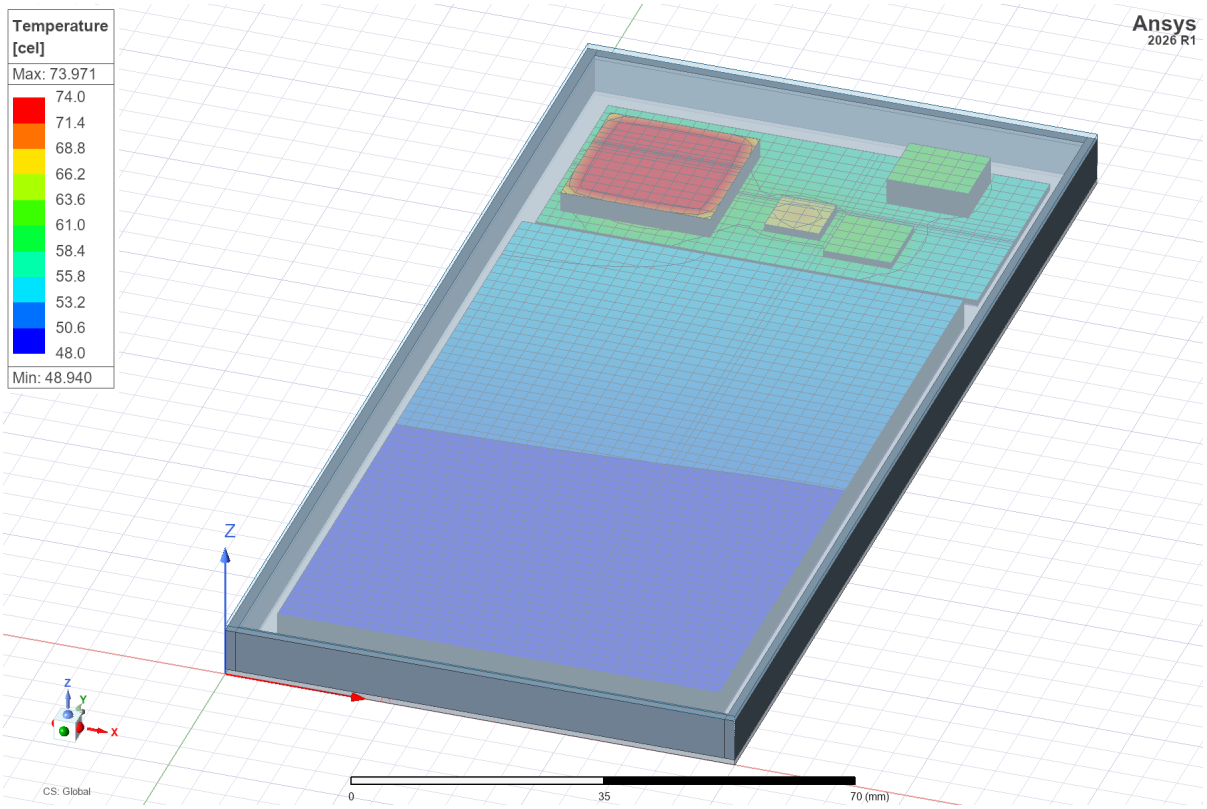


Figure 4. Fresh Icepak surface-mesh overlay on the battery and electronics. This shows visible surface elements, not a volume-quality or mesh-independence assessment.

Mesh item	Corrected setting or recorded output
Actual generated mesh	671,368 cells; 697,776 nodes
Requested maximum X/Y/Z sizes	2 / 2 / 0.05 mm
Requested minimum X/Y/Z gaps	0.05 / 0.05 / 0.01 mm
Minimum gap / edge elements	3 / 2
Multilevel meshing	Disabled
Mapping coverage	19 physical object names; 18 cooling face IDs

The 0.05 mm maximum Z request is smaller than the 0.2 mm PCB and processor TIM thicknesses. Requested sizes and mapping coverage support the repair, but do not establish element counts through each layer or acceptable local mesh quality.

Solver setup and convergence evidence

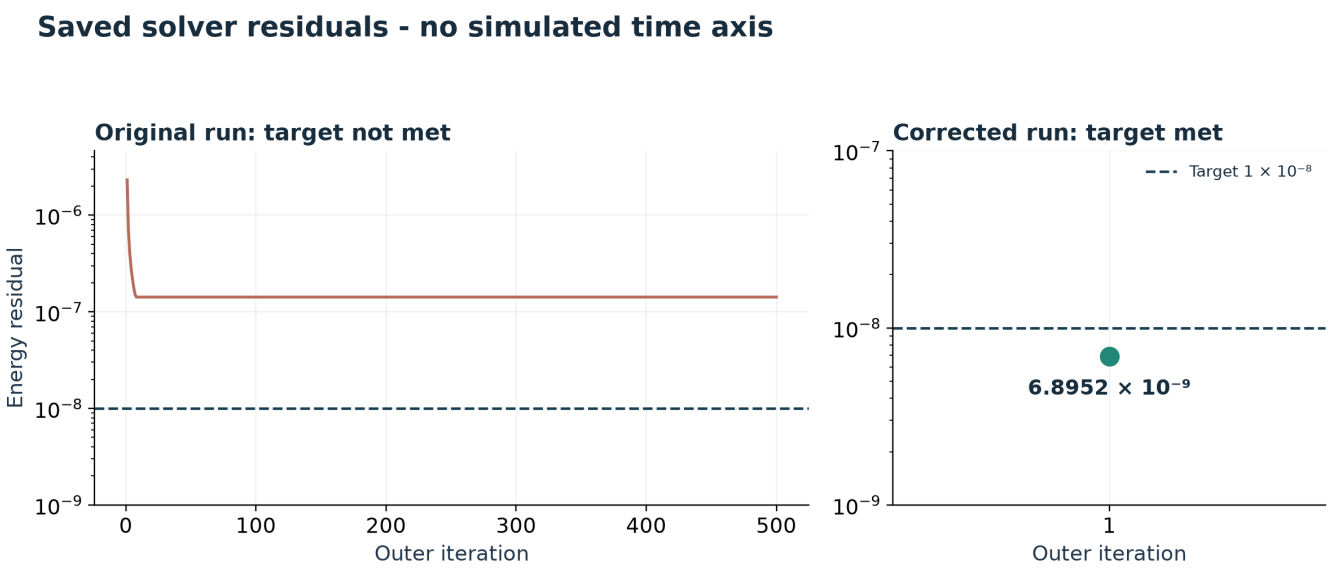


Figure 5. Recorded outer-iteration energy residuals. The corrected run has one saved iteration point; no time-dependent or extended convergence history is implied.

Solver item	Recorded value
Setup / mode	Phone_Steady_Corrected / SteadyState
Iteration limit / recorded outer iterations	1,000 / 1
Energy target / final residual	1×10^{-8} / 6.8952×10^{-9}
Initialization / temperature relaxation	25 C / 1
Runtime profile	4 cores; 63 seconds elapsed
Completion	Normal Completion at 20:20:29 PDT

A steady conduction solve can satisfy its energy equation in one outer iteration. The configured 1,000 value is a ceiling. The recorded residual meets its target, but residual convergence alone does not prove heat conservation, mesh independence or physical accuracy.

Both mesh and solve normal-completion entries are also returned by the native AEDT message API. The separate initialization message is preserved and discussed on page 12.

Component temperatures and statistics

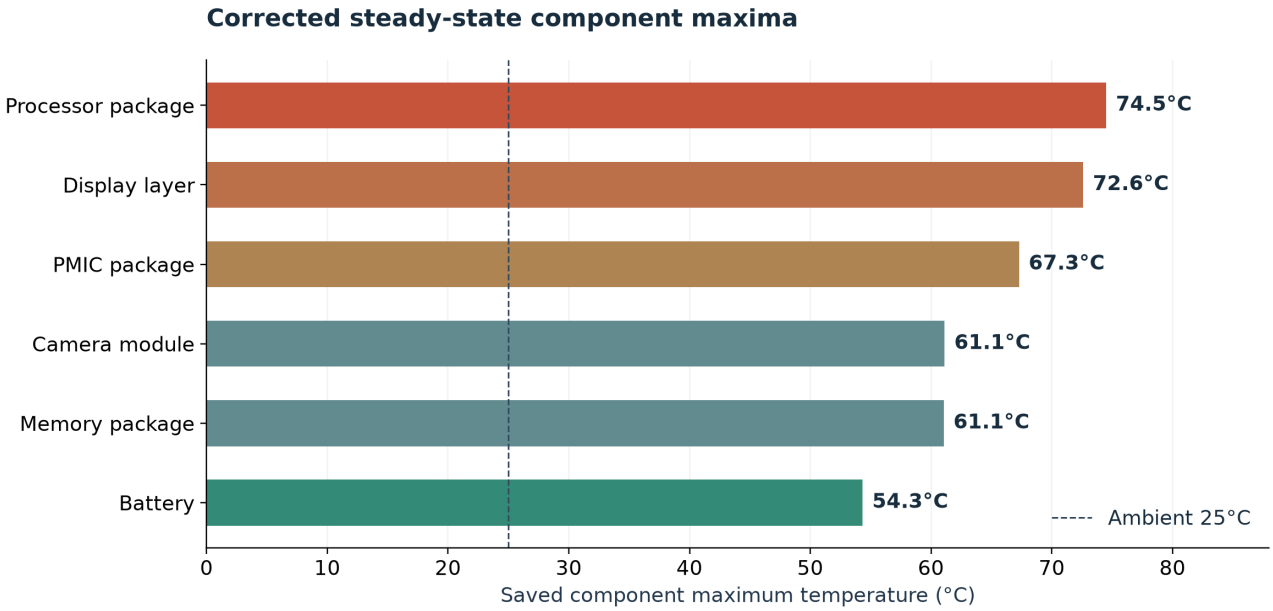


Figure 6. Saved volume maxima from DV68_S108_V132_137.SOV, converted from kelvin by subtracting 273.15.

Component	Saved maximum (C)	Vertex-average monitor (C)
Rear Cover	Not in saved SOV	51.43
Display Glass	Not in saved SOV	50.16
Display Layer	72.60	50.32
Battery Cell	54.33	50.67
Main PCB	Not in saved SOV	59.25
Processor Package	74.51	74.39
PMIC Package	67.29	66.74
Memory Package	61.06	60.39
Camera Module	61.13	60.54

Monitors are explicitly configured as **Vertex Average**. They are not volume averages. Processor package temperature is not semiconductor junction temperature. Unrecorded maxima have not been inferred from an average.

Battery temperature distribution

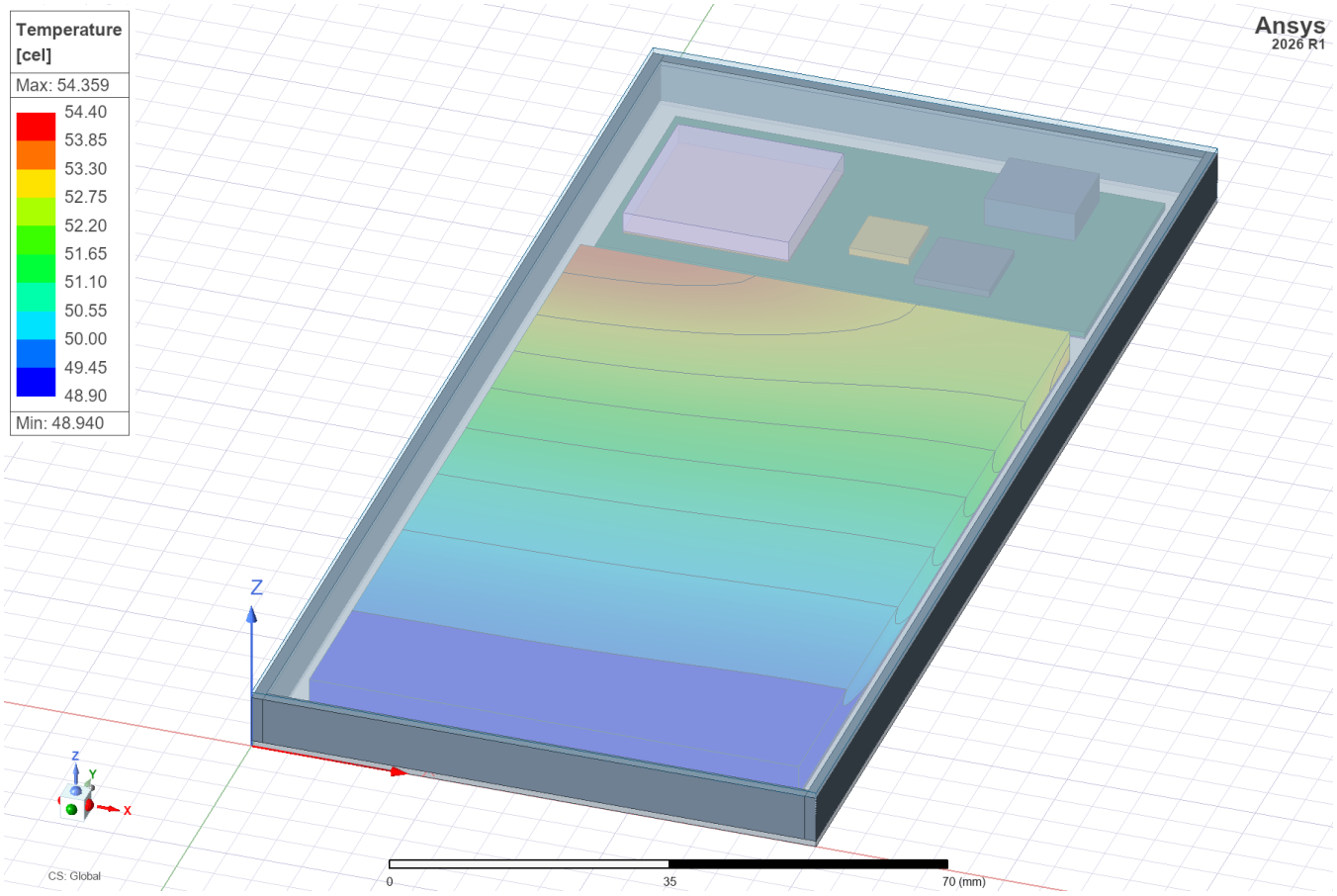


Figure 7. Native battery surface temperature contour. Display legend: approximately 48.940 to 54.359 C. The cooler region is near the lower end; the hotter region faces the electronics.

Battery metric	Value
Prescribed heat	0.8 W
Saved component volume maximum	54.33 C
Vertex-average temperature monitor	50.67 C
Native displayed surface maximum	54.36 C

The 0.03 C difference between displayed and saved maxima reflects the different surface-display and saved-volume evaluations. Do not use the contour legend as a replacement for the documented reporting statistic.

This is a fixed-heat thermal battery representation. A pass/fail conclusion requires the actual cell specification, operating conditions and a qualified model.

Electronics and enclosure hot spot

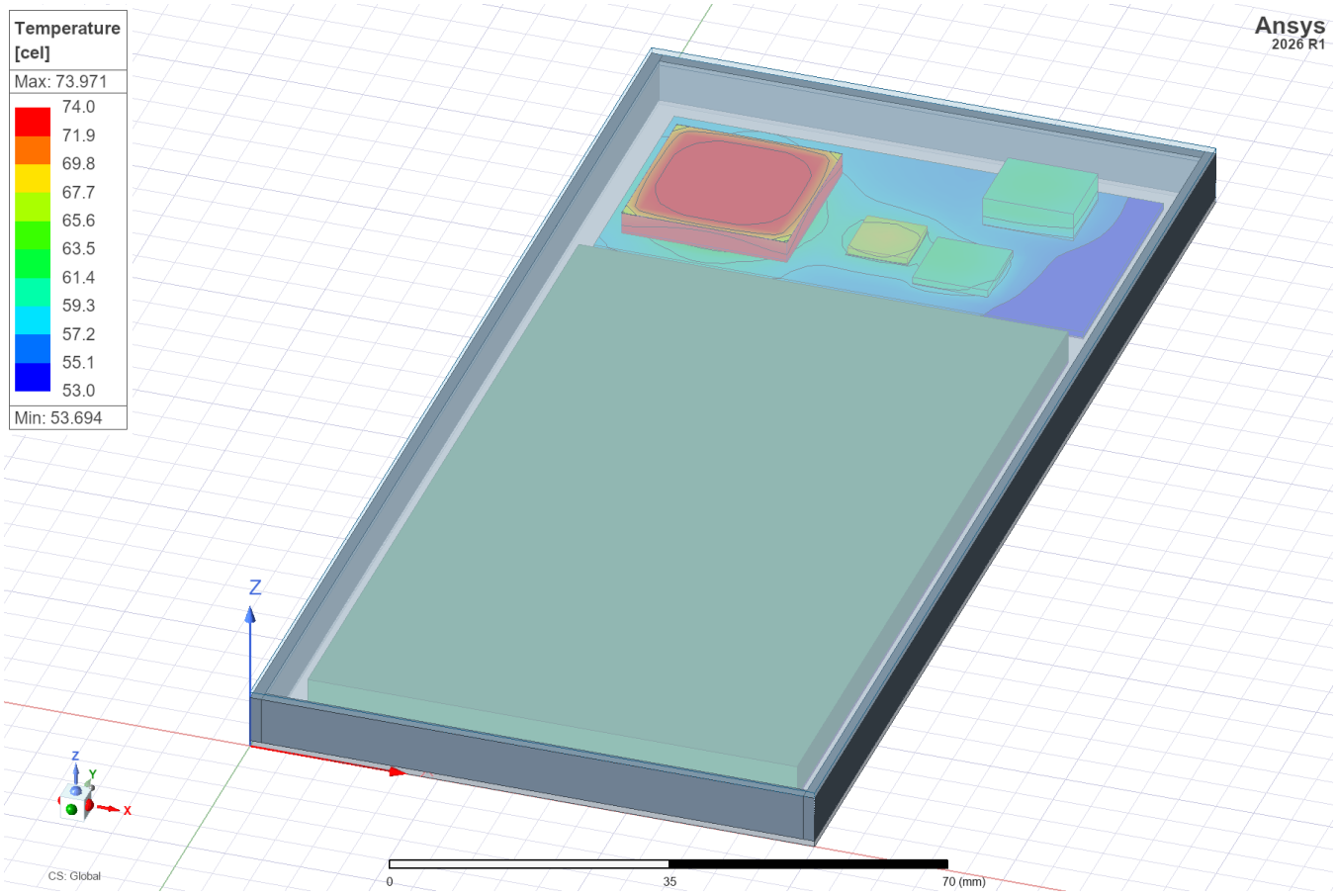


Figure 8. Native surface temperatures on the PCB, packages and processor spreading stack. The battery is context geometry in this view and has no temperature overlay here.

The saved processor package maximum is **74.51 C**. Its vertex-average monitor is **74.39 C**. The electronics selection in this surface view shows a maximum of **73.97 C**; the processor is partly obscured by the spreading stack, and the visible surface statistic is different.

The copper spreader and pad carry heat toward the display, producing a localized hot region over the processor. The PMIC, memory and camera have lower saved maxima: 67.29 C, 61.06 C and 61.13 C, respectively.

These are package and effective-material temperatures under assumed losses and contacts. The model does not resolve junction-to-case resistance or electrical temperature feedback.

Exterior surface temperature display

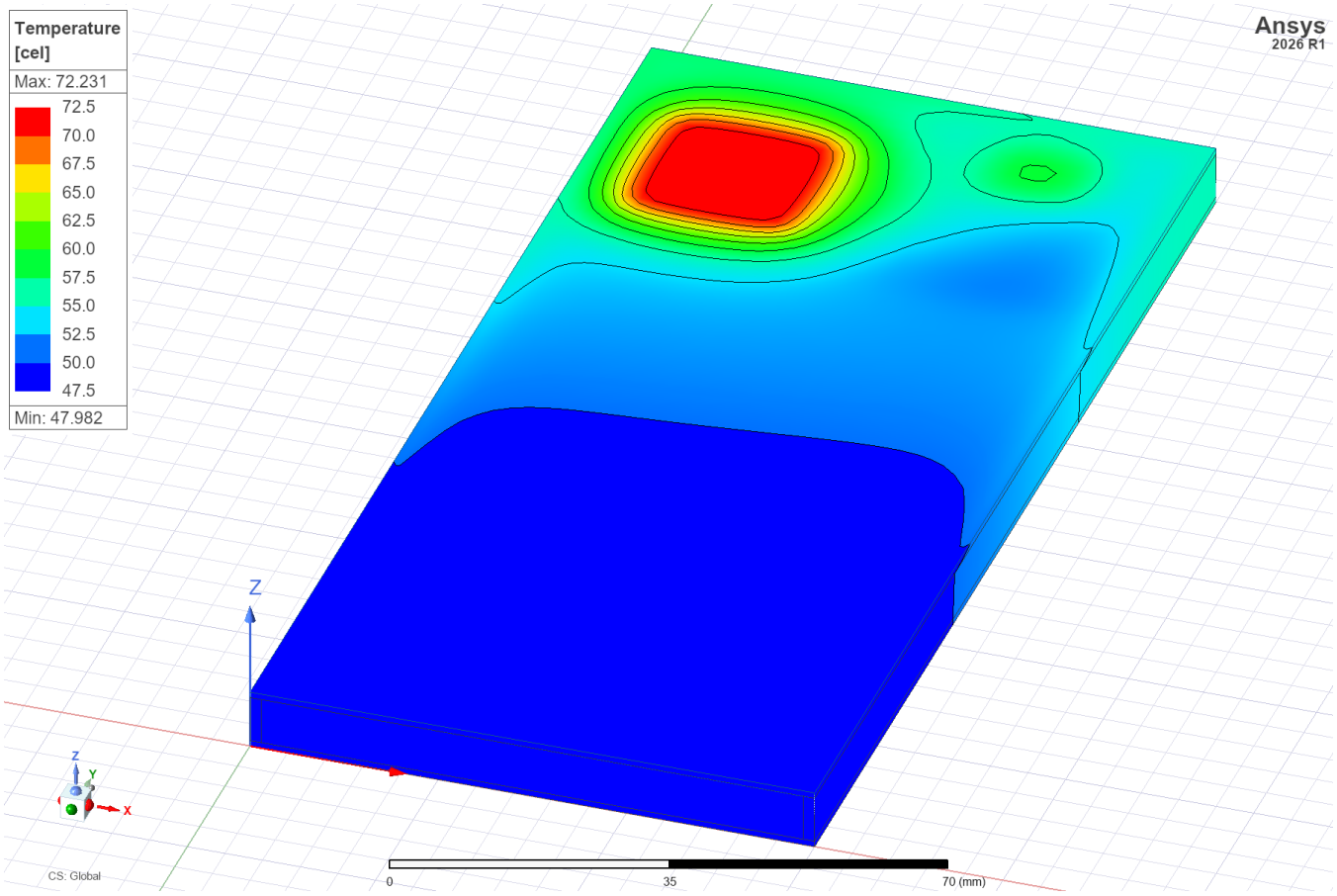


Figure 9. Native contour for the rear cover, frame and display glass selection. The visible front surface shows the processor-associated hot spot. Displayed selection range is approximately 47.98 to 72.23 C.

The selected-surface display reaches approximately **72.23 C**. This is a native displayed surface value, not a separately verified global skin maximum. The display-layer saved volume maximum is 72.60 C; display layer and display glass are different objects.

The rear-cover and display-glass vertex-average monitors are 51.43 C and 50.16 C. Those averages do not represent local contact temperatures or upper bounds.

The contour indicates where to focus an improved enclosure/contact model and hardware comparison. No temperature-limit, touch-temperature or product-compliance acceptance is made here.

What changed, and what remains open

Evidence	Original failed run	Corrected run
Cooling assignment	Region faces without usable cooling mesh	18 physical exterior solid faces
Thin layers	Missing mesh warnings for TIM/copper	All 19 parts appear in generated mapping
Mesh cells	25,876	671,368
Temperature limiting	Repeated cap at 5,000 K	No limiter messages
Energy residual	1.4195×10^{-7}	6.8952×10^{-9}
Interpretation	Invalid capped solution	Completed; qualification pending

This is a setup-repair comparison, not evidence of a physical cooling-design improvement. The old 4,726.85 C result was a numerical cap and must not be used to calculate a thermal-performance improvement.

Initialization diagnostic in corrected transcript

```
current.uns_out, lines 136-137:
Error: wta(1st) to string_length
Error Object: 0.
```

The solver continued, wrote fields and reported Normal Completion. The cause and effect of this initialization message remain unresolved; this report does not dismiss it as harmless.

A later reporting-only attempt returned "Surface mesh summary data is not supported". That occurred during report extraction, after the completed solve. The temperatures in this report therefore use the saved SOV and monitor data, rather than that unsupported summary export.

A transient solve is useful for warm-up duration and changing power. It does not repair a missing cooling path or replace qualification of a steady result.

Cooling comparisons: analytical screen only

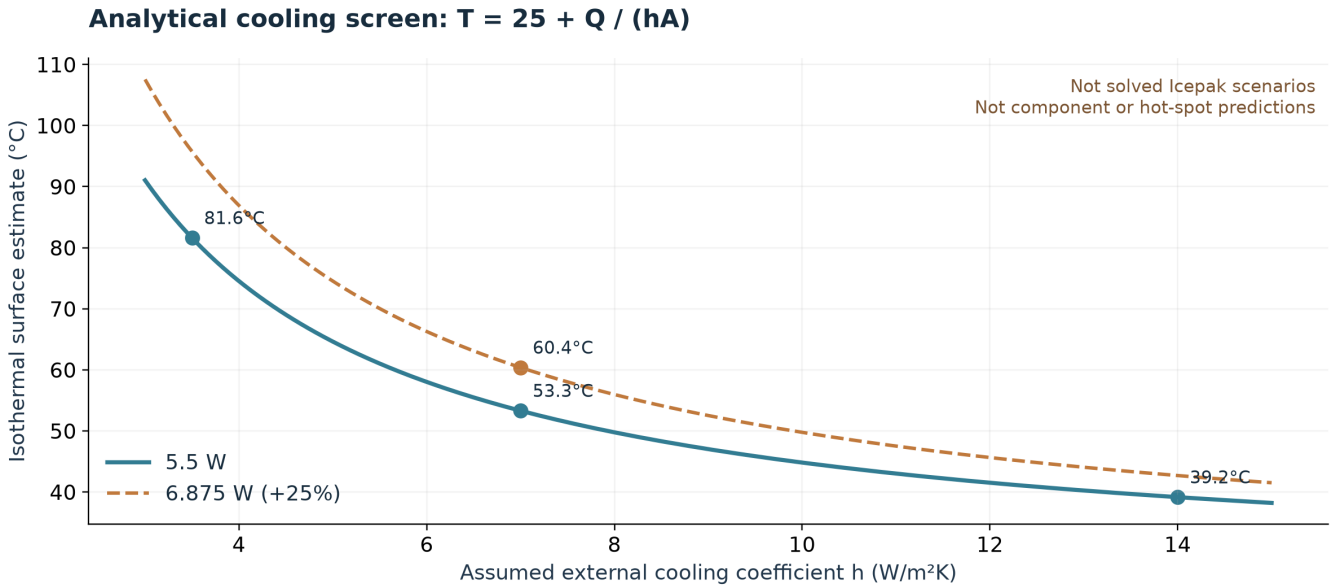


Figure 10. Independent Q/(hA) estimates using 0.02776 m2 external area and 25 C ambient. These are not new Icepak runs or predictions of battery/package maxima.

Analytical case	Heat (W)	h (W/m2 K)	Isothermal estimate (C)
Weaker cooling	5.5	3.5	81.6
Nominal assumption	5.5	7	53.3
Stronger cooling	5.5	14	39.2
Higher heat input	6.875	7	60.4

The estimate comes from Q = h A (T - ambient) with an isothermal surface. It omits internal thermal resistances and spatial hot spots. It is included only to show the influence of assumed cooling and total heat.

Recommended next solved comparisons: a refined baseline mesh; weaker and stronger external cooling at unchanged heat; a 25 percent heat increase; and, after numerical qualification, a transient power profile. No new simulations were started for this report.

Qualification and evidence register

Check	Current evidence / status
Execution	Mesh and solve both Normal Completion
Residual	Meets configured 1 x 10^{-8} energy target
Temperature limiting	None in corrected saved transcript
Cooling / thin-part mapping	18 cooling faces and 19 physical part names recorded
Initialization message	Open: investigate string_length error
Global heat balance	Open: independently compare 5.5 W input with outward heat
Local mesh quality / refinement	Open: local layer coverage and refined-grid comparison
Physical validation	Open: replace illustrative assumptions and compare with measurements

Primary evidence in the package

Evidence key	File or source
A	DV68_S108_V112.profile: actual mesh totals and normal completions
B	DV68_S108_V131.restart/current.uns_out: solver transcript
C	DV68_S108_V132_137.SOV: six component volume maxima
D	DV68_S108_MON0..9: residual and vertex-average monitors
E	DV68_S108_Meshes0_V0.sd/grid_mapping: parts and boundaries
F	original_inputs_unchanged.json and exterior_cooling_face_audit.json
G	Fresh native images, live_native_audit.json and source hash manifest

The near-zero per-source heat-transfer entries in the SOV are not a verified global outward heat balance. The selected coupled/shadow surfaces are not used here to claim conservation.

Reopen results and use the transfer package

In Ansys Electronics Desktop

1. Open Illustrative_Phone_Corrected.aedt with its matching .aedtresults folder beside it. Use the full native archive for saved field access.

2. Expand IcepakDesign1 > Analysis > Phone_Steady_Corrected. Inspect its profile and convergence; the saved result is SteadyState.

3. Select Battery_Cell, then Icepak > Fields > Plot Fields > Temperature. Choose Phone_Steady_Corrected : SteadyState and surface display.

4. Repeat with the PCB/packages and with the enclosure/display glass. Read each legend and selection carefully; surface-display maxima differ from saved volume maxima.

5. For engineering acceptance, complete the heat-balance, initialization-diagnostic and refinement checks before interpreting design margins.

On the Mac

Download the transferred ZIP from the private Drive link, unzip it, and open this PDF or index.html. The HTML uses only relative local image links. PNG images, SVG charts, CSV tables and report_facts.json can be reused in the PCB webpage.

The compact package contains the report, images, inputs and small evidence files. The full native archive additionally includes the matching .aedtresults tree. Reviewing the PDF/HTML on the Mac does not require Ansys.

```
Project: Illustrative_Phone_Corrected
Design:  IcepakDesign1
Setup:   Phone_Steady_Corrected
Result:  SteadyState
Source folder: repair_20261018_193055
```

Historical ready_configuration_audit.json and repair_status.json predate the user-performed successful run. They are retained in the full snapshot as history. Use the later profile, SOV, transcript and this report_facts.json for current execution status.